Userspace space OpenAMP application uses shared memory for communicating with the OpenAMP based shared devices running in RPU or **MicroBlaze V**. The default Ubuntu kernel does not allow the userspace application to access the shared memory. So following steps are followed to update the Kria Ubuntu kernel and build in the development Ubuntu machine.

Install dependency packages

- Install packages required for build

```
sudo apt-get install git fakeroot libncurses-dev gcc-aarch64-linux-gnu linux-tools-common
```

- Update ubuntu source list :

```
echo "deb-src http://archive.ubuntu.com/ubuntu jammy main" | sudo tee -a
/etc/apt/sources.list.d/jammy..list
sudo apt-get update
```

- Next install build dependencies for kernel

```
sudo apt-get build-dep linux
```

Clone the Linux Kernel Source Code

- Get the Linux kernel source code for Kria Ubuntu

```
git clone
https://git.launchpad.net/~canonical-kernel/ubuntu/+source/linux-xilinx-zynqmp/+git/ja
mmy
```

After download is finished, checkout to the branch Ubuntu-xilinx-zynqmp-5.15.0-1027.31 corresponding to the current linux build.

```
cd jammy
git checkout Ubuntu-xilinx-zynqmp-5.15.0-1027.31
```

Configure the Kernel

Run following commands to get the config console for the kernel :

```
export ARCH=arm64
export CROSS_COMPILE=aarch64-linux-gnu-
fakeroot debian/rules clean
```

Next give custom name for the current kernel build by updating debian.zynqmp/changelog file
Open the file using nano editor :

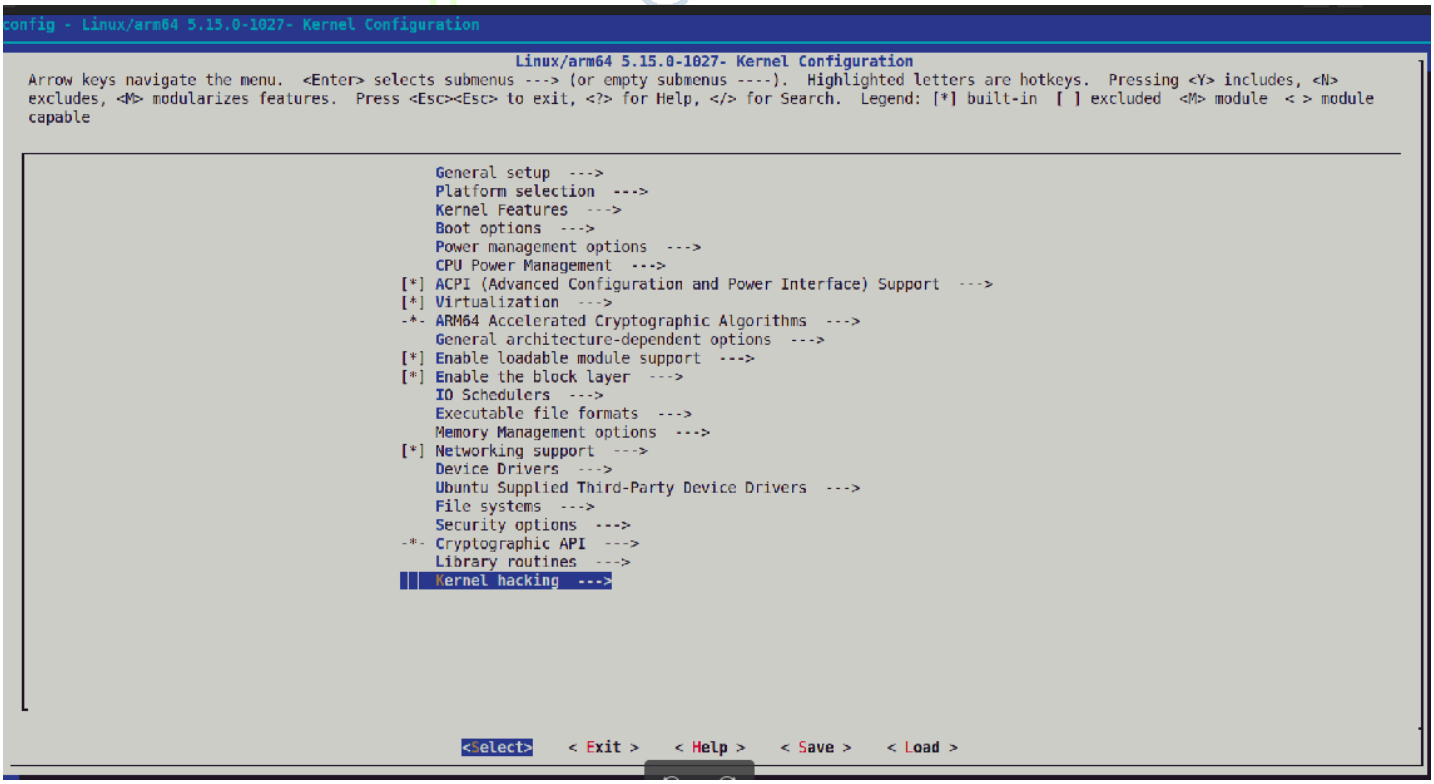
```
nano debian.zynqmp/changelog
```

And update the first line with :

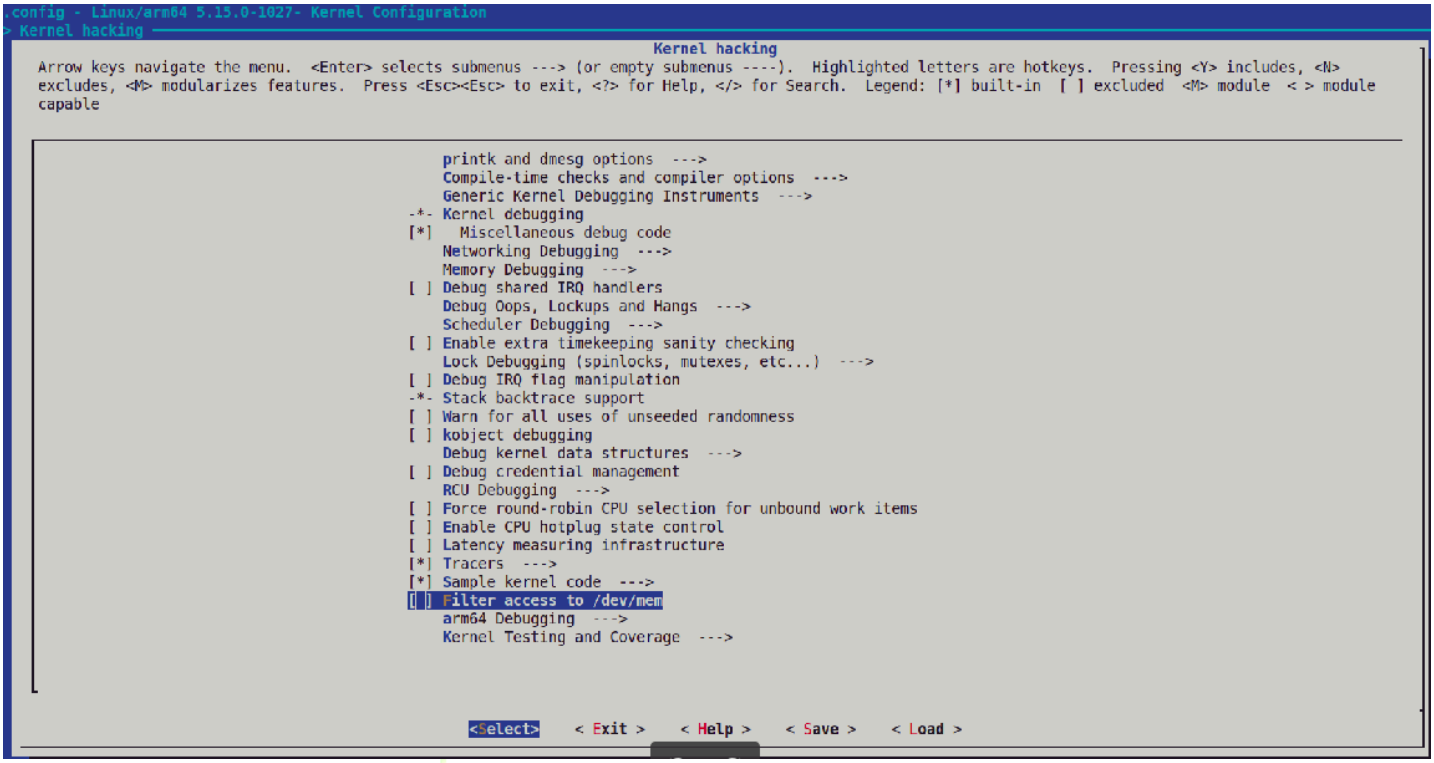
```
linux-xilinx-zynqmp (5.15.0-1027.31+local) jammy; urgency=medium
```

Next run following command to get the kernel config interface :
fakeroot debian/rules clean
fakeroot debian/rules editconfigs

This will open the kernel config interface:



Disable the “Filter access to /dev/mem” corresponding to “STRICT_DEVMEM” kernel config in “Kernel hacking” option



Exit and save the kernel config.

Build the kernel

Next build the linux kernel by running following commands in the same terminal with environment variable set for arm64

```
export DEBIAN_KERNEL_DISABLE_CHECKS=yes
fakeroot debian/rules clean
do_tools=false fakeroot debian/rules binary
```

This will build the kernel and create linux deb packages(linux-xxxx--5.15.0-1027-xilinx-zynqmp_5.15.0-1027.31+local_arm64) just one folder outside jammy folder.

Installing the linux kernel

To install the above build linux kernel packages, copy the linux kernel to the board and install using dpkg command :

```
sudo dpkg -i *.deb
```